

SEVEN PERSONS AGDM, AB, Canada

WMO: 715250

Lat: 49.9183N

Lon: 110.9149W

Elev: 2439

StdP: 13.45

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.9	-13.2	-25.0	1.5	-18.9	-19.0	2.1	-13.3	31.1	32.9	27.9	29.0	7.4	230	0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	90.4	62.8	86.8	62.0	83.0	61.2	66.5	82.1	64.6	80.5	63.1	78.8	9.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	88.9	71.8	59.1	82.0	70.1	57.1	76.2	68.5	32.5	81.9	31.0	80.7	29.8	78.7	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	24.7	21.5	DB	-28.5	96.3	4.6	3.1	-31.8	98.6	-34.5	100.4	-37.1	102.1	-40.5	104.4	(4)
(5)			WB	-28.6	70.6	4.7	2.0	-32.0	72.0	-34.7	73.2	-37.3	74.4	-40.7	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	19.3	18.0	31.0	42.4	52.7	60.1	66.9	64.9	55.8	43.1	29.8	19.2	(6)	
	DBStd	20.59	15.45	15.25	13.49	8.96	7.93	6.04	5.72	6.45	8.50	9.42	13.22	14.27	(7)	
	HDD50	4606	952	896	591	252	62	3	0	0	36	249	609	956	(8)	
	HDD65	8604	1417	1316	1054	677	389	172	46	83	296	678	1055	1421	(9)	
	CDD50	1711	0	0	2	24	145	306	523	462	209	36	4	0	(10)	
	CDD65	234	0	0	0	0	6	25	104	80	19	0	0	0	(11)	
	CDH74	4027	0	0	0	15	191	449	1626	1301	430	15	0	0	(12)	
	CDH80	1459	0	0	0	2	40	125	639	501	151	1	0	0	(13)	
(14) Wind	WSAvg	10.5	11.8	10.4	11.0	11.5	10.7	10.0	8.8	8.8	9.6	10.8	11.2	11.0	(14)	
(15) Precipitation	PrecAvg	15.4	0.8	0.6	0.8	1.4	2.2	3.1	1.5	1.3	1.3	0.9	0.7	0.7	(15)	
	PrecMax	23.3	1.5	1.5	1.5	2.6	3.7	6.8	5.7	3.5	2.5	2.3	1.7	1.5	(16)	
	PrecMin	9.1	0.2	0.1	0.1	0.4	0.4	0.7	0.2	0.3	0.0	0.2	0.2	0.1	(17)	
	PrecStd	2.8	0.3	0.4	0.4	0.6	0.9	1.5	1.1	0.8	0.7	0.5	0.4	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	53.5	65.2	75.4	85.2	89.4	95.0	93.3	90.3	76.0	63.8	50.0	(19)	
		MCWB	42.6	41.9	48.6	53.0	59.3	63.8	65.0	62.2	60.2	56.1	48.1	40.4	(20)	
	2%	DB	45.1	46.2	59.6	69.3	79.2	83.3	90.9	89.7	84.8	68.3	56.1	44.7	(21)	
		MCWB	38.6	38.2	45.5	50.5	56.6	61.7	64.1	61.7	58.7	52.2	45.0	37.8	(22)	
	5%	DB	41.3	41.0	54.1	64.4	74.6	79.1	87.0	86.0	78.8	63.3	51.1	41.3	(23)	
		MCWB	36.3	35.1	42.6	47.8	55.0	60.4	63.6	61.4	57.2	49.5	42.0	36.0	(24)	
	10%	DB	38.5	37.4	49.0	59.5	69.9	75.1	83.1	81.6	73.1	58.6	46.6	37.9	(25)	
		MCWB	34.6	33.4	39.9	44.9	52.4	58.5	62.8	60.5	54.8	47.0	39.0	33.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	42.9	49.3	54.2	60.4	67.4	70.2	67.2	63.2	56.5	50.0	41.8	(27)	
		MCDB	49.6	50.4	63.2	72.5	80.0	80.9	86.5	82.2	80.8	73.6	59.8	47.6	(28)	
	2%	WB	39.0	38.5	46.4	51.1	58.1	64.3	67.5	64.7	60.4	53.1	45.6	38.2	(29)	
		MCDB	44.6	45.8	57.9	67.9	75.8	79.4	83.1	81.7	80.7	65.9	55.4	43.2	(30)	
	5%	WB	36.6	35.5	43.4	48.3	56.1	62.0	65.6	63.2	58.1	50.2	42.3	35.9	(31)	
		MCDB	40.7	40.3	53.3	62.2	71.5	75.9	81.2	80.7	76.7	61.5	49.8	40.8	(32)	
	10%	WB	34.6	33.4	40.2	45.8	54.0	59.6	64.1	61.5	56.0	47.8	39.5	33.4	(33)	
		MCDB	38.4	37.5	48.3	57.5	67.2	72.3	79.3	78.2	72.3	57.8	45.6	37.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	19.3	20.5	23.5	24.5	23.8	27.9	27.6	25.9	22.8	20.0	17.4	(35)	
		MCDBR	18.2	22.5	28.0	33.6	32.7	30.6	34.0	35.3	35.9	29.4	25.7	19.5	(36)	
	5% WB	MCWBR	13.4	15.5	16.0	17.3	14.4	13.5	13.5	13.5	14.7	15.1	15.4	14.1	(37)	
		MCWBR	17.7	21.0	27.1	31.2	29.0	28.2	29.7	30.7	33.7	28.0	24.0	19.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.271	0.271	0.300	0.356	0.379	0.369	0.368	0.390	0.343	0.301	0.269	0.258	(40)	
		taud	2.393	2.463	2.443	2.285	2.264	2.353	2.372	2.278	2.392	2.490	2.482	2.405	(41)	
	Ebn at Noon	Ebn at Noon	250	283	290	281	279	282	280	267	270	266	251	239	(42)	
		Edh at Noon	20	24	29	38	41	38	36	38	30	23	18	17	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	395	709	1133	1567	1857	2030	2207	1777	1268	787	441	308	(44)	
	RadStd	RadStd	37	59	79	109	132	120	110	113	104	79	31	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page